

Table 2

Information about the datasets used in this paper. PARARULES has less number of examples that require deep reasoning steps. CONCEPTRULES V2 does not consider reasoning depths greater than 3. The train, dev and test set are already splitted by the author of each dataset.

Dataset		Depth=0	Depth=1	Depth=2	Depth=3	Depth=4	Depth=5
PARARULES	Train	290435	157440	75131	48010	9443	7325
	Dev	41559	22276	10833	6959	1334	1038
	Test	83119	45067	21496	13741	2691	2086
PARARULE-Plus	Train	-	-	89952	90016	90010	90022
	Dev	-	-	16204	16154	16150	16150
	Test	-	-	2708	2694	2704	2692
CONCEPTRULES V2 (full)	Train	2074360	1310622	873748	436874	-	-
	Dev	115148	72810	48540	24270	-	-
	Test	115468	72810	48540	24270	-	-
CONCEPTRULES V2 (simplified)	Train	131646	74136	49424	24712	-	-
	Dev	7166	4116	2744	1372	-	-
	Test	7362	4116	2744	1372	-	-

imbalance, we develop a large dataset on multi-step reasoning over natural language called PARARULE-Plus that has a balanced distribution over different reasoning depths. The test accuracy on deeper depths and extra out-of-distribution examples is greatly improved when we add PARARULE-Plus in the training process (Table 7 and 8). The experiment result also verifies the necessity of our dataset.

2. Related Work

Systems that integrate deep learning techniques with symbolic reasoning are called *neuro-symbolic systems* [8]. Many such neural reasoning models can be viewed as logical program interpreters. Neural-symbolic machines (NSM) [9] and neural program interpreters (NPI) [10] are such examples. NSM describes a framework that consists of a seq-to-seq neural programmer, a Lisp interpreter to execute the program, and iterative maximum likelihood to train the model; NPI presents an RNN to learn and represent logic programs. NPI is designed for compositional programs, including addition, sorting, and canonicalising 3D models. Reinforcement learning has also been applied to learn Prolog-like algorithms [11]. Distributed representations of predicates and constants for traditional symbolic reasoning engines can be learned by neural theorem provers [12]. In our method, end-to-end neural networks learn representations at word level and learn to reason with natural language.

Several reasoning datasets have been introduced for natural language-based reasoning tasks which can be used to evaluate and compare neural models’ reasoning capabilities. Roughly speaking, the datasets can be categorised as “shallow reasoning” and “deep reasoning” tasks. The first category includes Task 15 in the bAbI dataset v1.0 [4], conditional probes in [13], and “multi-hop” reasoning dataset HotpotQA [14]. In these datasets, the reasoning instances usually do not go beyond 2-steps. A typical example in the bAbI dataset would be “Mouse is

afraid of cats. Alice is a mouse. What is Alice afraid of? A: cats.” and a typical example in the conditional probes of [13] is “If A has visited B, then C has visited D. A visited B. Has C visited D? A: Yes.” The main difference of HotpotQA with the other two is that the rules in HotpotQA are embedded in sentences. In HotpotQA, a sentence contains both factual information and rule information. Figure 1 shows a 2-hop example from HotpotQA dataset.

The second category contains instances where the reasoning depth may go beyond 2. These datasets are PARARULES [3], CONCEPTRULES V1 [6] and CONCEPTRULES V2 [7]. Examples in PARARULES may require reasoning depth as deep as 5, while CONCEPTRULES V1 and V2 require depths as deep as 3. PARARULES differs from the three above in the following senses: First, solving the problems in bAbI Task 15 requires implicit rules. For example, “Alice goes to the park. Peter goes to the restaurant. Where is Alice? A: park” requires the rule “A moves to B $\rightarrow$ A at B”. Contrary to the bAbI tasks, PARARULES requires reasoning with explicit rules that is more akin to logic programming. Contrary to the HotpotQA dataset, in PARARULES, the facts and rules are separate. One of the main issues of PARARULES, CONCEPTRULES V1 and CONCEPTRULES V2 is the unbalanced distribution over reasoning depths. The datasets have more examples of shallow reasoning (Depth<2) than that of deep reasoning (Depth $\geq$ 2).

Other work uses Transformer-based pretrained language models to perform tasks that require multi-step reasoning over natural language. One such work is [3] which demonstrated that the pretrained language models (RoBERTa [5], and BERT [15]) can be used for natural language-based reasoning tasks. However, it is unknown whether these pretrained language models would perform better than neural networks specifically designed for reasoning or fact-checking tasks [16, 17, 18, 19, 20, 17, 21, 22]. Attention mechanisms have been used to enable efficient inference by shortening input lengths, thereby improving text generation and classification performance [23]. Previous studies have shown that data augmentation can enhance model performance on logical reasoning tasks and knowledge-based question answering [24, 25, 26, 27, 28, 29, 30, 31, 16, 28]. One of our goals is to compare them against models based on iterative memory mechanisms which have been shown to perform well on multi-step reasoning tasks over logic programs.

(Paragraph A:) **LeBron James** won **the 2015-2016 NBA Championship**.
 (Paragraph B:) **LeBron James** is a basketball player for **Cleveland Cavaliers**.
 (Question:) Which team did the players who won the 2015-2016 NBA Championship play for? (Answer:) Cleveland Cavaliers.

Figure 1: A depth-2 example from HotpotQA [14].

3. Problem Definition

We consider multi-step deductive reasoning over natural language. Each sample is a triple (*Context*, *Question*, *Answer*) where *Context* contains natural language implications (rules) and observations (facts) resembling a knowledge base in logic programs, *Question* is a natural language sentence expressing an atomic fact, and *Answer* $\in$ {true, false} tells whether *Question* naturally follows from *Context*. In this regard, (*Context*, *Question*, *Answer*) is a

natural-language counterpart to a logic program. Figure 2 illustrates several examples in the PARARULES dataset [3]. The rules are expressed in phrases such as “If A, then B” and “All A are B”, and facts are represented using propositions such as “A is happy” and “B is funny”.

Multi-step reasoning requires multiple reasoning steps to answer a question. We define the reasoning *depths* (or *steps*) as the number of rules required to answer a question. For example, from “Bob is smart.”, it takes only one rule “All smart people are talented.” to answer “Bob is talented?”. This example is therefore considered as depth-1 reasoning. PARARULES has seven sub-datasets, each is named by the greatest depth of reasoning required to deduce which related facts support its question: depths $D = 1$, $D = 2$, $D = 3$, $D \leq 3$, $D \leq 3 + \text{NatLang}$, $D \leq 5$, $D \leq 5 + \text{NatLang}$, respectively. Here, NatLang means the extra out-of-distribution examples, containing about 2,000 examples. These examples include questions of different depths. They were created by paraphrasing examples using crowdsourcing. Crowdworkers rewrite part of the synthetic dataset using phrases such as “often”, “rather resembles”, and “a bit”. For instance, sentences like “Charlie is green, but *often* kind” and “Harry *seems* to be round” are more natural to a human reader.

(Input Facts:) Anne is rough. Anne is blue.
 (Input Rules:) Rule 1: Cold people are rough.
 Rule 2: Rough people are young.
 Rule 3: If Anne is green then Anne is blue.
 Rule 4: If someone is rough and nice then they are green.
 Rule 5: If someone is rough and furry then they are blue.
 Rule 6: All young people are cold.
 Q1: Anne is cold. True/False? [Answer: T]
 Q2: Anne is not young. True/False? [Answer: F]
 Q3: Anne is not green. True/False? [Answer: T]

Figure 2: Examples from PARARULES [3]. The context (facts + rules) and the question are grouped as the input, and the output is a Boolean value indicating if the question is true or false, given the context.

While PARARULES separates fact and rule explicitly, CONCEPTRULES V1 and V2 [6, 7] put fact and rule together. Each of CONCEPTRULES V1 and V2 has a simplified and a full version. Both CONCEPTRULES V2 (simplified) and CONCEPTRULES V2 (full) include *negation as failure* (NAF) and derivable cases. Derivable means answers can be derived from context and question. Negation as failure means if we cannot find facts or rules to derive the answer, we assume it is false. The different versions of CONCEPTRULES are summarised in Table 3. In CONCEPTRULES V2 (full), the rulesets are randomly shuffled and random textual noise is added. Both CONCEPTRULES V1 and V2 contain examples with reasoning depths from 0 to 3. CONCEPTRULES V1 does not label the reasoning depth for each example, while CONCEPTRULES V2 contains the labels.

4. Method

This section describes a word-level RNN-based iterative neural network. The general idea of the model is borrowed from DeepLogic. DeepLogic is an end-to-end iterative memory attention